4. Protocol State Awareness

4.1 Multi-Layer Message Analysis

Device Endpoint Analysis → Message Format Description Extraction → LLM-based Message Format Generation

Cluster	clusterID	scenario
In-Cluster		
Basic	0x0000	HA
Out-Cluster		
OTA Upgrade	0x0019	HA
MS-Cluster		
Unkown1	0xe000	HA

Supported Clusters

Message Type ⊕ Message Format (JSON)

4.2 Message Relationship Analysis

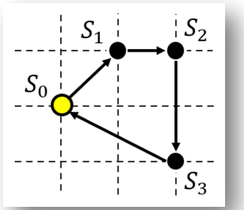
Message Dependency Collection ⊕ Message Correlation Discovery

Dependency & Correlation Message Group

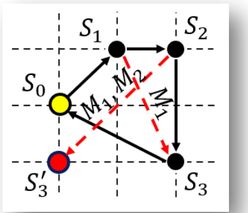
4.3 Protocol State Space Exploration

State Graph Construction → Potential State Transition Discovery

Protocol State Graph



Update



Fuzzing Graph

5. Device State Awareness

5.1 State Attribute Collection

Stage 1: Active Query

Stage 2: Specification Library Analysis → Development Documentation Analysis

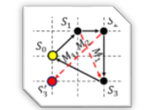
Stage 3: Hidden State Discovery

Complex Data Type Parse

5.2 Attribute Writability Analysis

Attribute	Data Type	Writability
Hidden State Attributes		
GroupCount	int16	[Add, Remove, …]
Basic State Attributes		
StartUpColor Temperature	uint16	WriteAttributes

6. State-Guided Fuzzer



Attribute	Data Type	Writability
Hidden State Attributes		
GroupCount	int16	[Add, Remove, …]
Basic State Attributes		
StartUpColor Temperature	uint16	WriteAttributes

Protocol Sate ⊕ Device Sate

JSON → Test Case Generator

1 **SET** Device State Message

2 **SEND** Mutated Message Sequence

Real Testbed

Device Watchdog

Crash

Abnormal State